

3rd Year (BSc) Project Handbook

This document contains general information about the 3rd year project course COMP3003. For more information, you should attend the lectures and read the recommended textbook: *C. Dawson, The essence of computing projects: A student's guide*.

Please refer to the Moodle COMP3003 webpage for any latest updates.

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Course Descriptions

- Please enter [NottinghamHub - COMP3003 Individual Dissertation Single Honours](#) to see the description.

WARNING: If you are a 2+2 student and want to obtain your graduation certificate from Chinese Ministry of Education (MOE), then you **must** take the Individual Dissertation course. This course is compulsory for you.

AI Use Policy

AI tools must not be used to generate content or code from scratch or to replace original academic work.

Any use of AI tools must fully comply with University plagiarism and academic integrity policies, including proper acknowledgment where required. Students are responsible for structuring their dissertations appropriately, expressing all ideas in their own words and carrying out the technical work for their projects.

[NEW] To ensure clarity and accountability, students must include an explicit **declaration of AI usage** at the end of their dissertation.

It is extremely important that students fully understand their own work, including any code submitted as part of their dissertation, even when AI tools have been used for support. Students should be able to explain, justify, and defend every aspect of their work to their supervisors at all times throughout the academic year, and to the markers during demonstrations. Any use of AI-supported work that is not appropriately acknowledged, or any work that a student is unable to adequately explain and defend, may be considered evidence of academic misconduct and will be dealt with in accordance with the University's regulations.

Course Accreditation – British Computer Society

Some programmes have, or have been recommended for, [British Computer Society](#) (BCS) accreditation for Chartered IT Professional (CITP) and in some cases partial fulfilment for Chartered Engineer (CEng) or Chartered Scientist (CSci). It is expected that within an undergraduate programme, you will undertake a major computing project, normally in final year and normally as an individual activity, to meet the BCS accreditation requirements, giving you the opportunity to demonstrate:

- ability to apply practical and analytical skills present in the programme as a whole
- innovation and/or creativity
- synthesis of information, ideas and practices to provide a quality solution together with an evaluation of that solution
- that the project meets a real need in a wider context
- ability to self-manage a significant piece of work
- critical self-evaluation of the process

Projects must also be passed with a minimum mark of 40%. For further information, please refer to the [British Computer Society Guideline](#).

The BCS website also provides an overview of the [benefits of accreditation](#).

People Involved: Roles and Responsibilities

The Student

You are the key person who has the full responsibility for every aspect of your project. It is likely to be the most difficult yet gratifying single piece of work that you have undertaken so far. It is essential to realise that it is **your** project: success or failure is particularly up to you. You will receive guidance from your supervisor and there are other sources of help available, including this project handbook, Moodle module web pages, lectures, and more. Nevertheless, **you** will plan the project, manage it and execute the tasks. You should be pro-active and constructive in your approach.

You will be expected to handle the issues that you face (it is very likely that you will due to nature of projects) and resolve them by finding appropriate solutions. It is **not** your supervisor's duty to find solutions to the problems for you, although they will provide guidance where appropriate. How you cope with the challenges that you encounter throughout the project will determine how much you learn and will be reflected in your overall mark. You will also get significant satisfaction from having completed your project overcoming issues along the way.

A Summary of Key Student Responsibilities

Project selection:

- Agree on a project with a supervisor well before the start of the final year. There may be some circumstances where this is not possible, in which case you need to find a project and a supervisor without delay. The list of supervisors and projects that they offer can be reached from the Moodle [Project Allocation System](#) (PAS). Please remember the *expectation* that "Students undertake a project in Computer Science that is relevant to their programme of study; for example, projects undertaken by Artificial Intelligence (AI) students must have a strong AI focus.". Projects based solely on literature review activity and/or user/market surveys are not acceptable.

External Sponsor:

- This is not common on L3 final-year projects. However, if there is an *external sponsor* (e.g., company) involved in your project, it is your duty to inform the Project Coordinator by sending a brief explanation of the project via email. You should also ensure that the external sponsor is aware of the confidentiality and IP issues and you have an appropriate agreement before taking on the project. Collaborative work with an external sponsor could require confidentiality and/or an IP agreement. Please read the section on Intellectual Property Rights to know your rights.

Project Proposal:

- As your first deliverable, write the Project Proposal
- Get it approved by the supervisor.

Research Ethics and Data Management Plan:

- **Undergraduate projects are not expected to require full ethics clearance.** All project studies/research should be designed to involve **minimal risk**. If, for any reason, a project is identified as requiring full ethics clearance, the supervisor should work with the student to revise the project scope so that it remains minimal risk. You can refer to the [SCS-REC review code of practice](#) for further details on ethics forms and examples. **You should use and submit the appropriate [Standard Operating Procedure \(SOPs\)](#) provided in the SharePoint site.** These are partially pre-completed ethics application forms for specific types of project. Please refer to the ethics clearance submission instructions provided on the Moodle project webpage.
- All research projects should have a Data Management Plan (DMP) and you can refer to the SharePoint site for the [DMP form and guidance](#).

Remember that it is part of project management to have these documents signed off by the deadline. Ethical review and approval is required before the study begins for all projects involving (i) the participation of human subjects, or (ii) the use of personal data or, (iii) biological material. If a project requires full ethics clearance **and is not considered minimal risk**, the ethics approval process, including the preparation of the ethics review application, should be led by the supervisor.

The work done without appropriate ethics approval is an academic offence/misconduct.

[The University of Nottingham's Code of Research Conduct and Research Ethics](#) provides a comprehensive framework for good research conduct according to the appropriate ethical, legal, and professional frameworks and standards.

Project Management:

- Prepare a project plan and work to it.
- Manage your own time effectively.

Unlike your taught modules, this is all your own time to spend on the module: the project is therefore significantly larger in scale than any piece of assessed coursework.

Requesting Hardware and Software:

If you need special hardware and software for your project and your supervisor agrees, you will need to submit a request through <https://selfservice.nottingham.ac.uk/>

Supervision:

- Agree with your supervisor how you will get in touch to arrange supervision.
- Attend supervision meetings.
- Check your **UoN** email and Moodle announcements of the project module regularly.

Deadlines:

- Meet the final hand-in deadline and interim deadlines (e.g., for completing the Project Proposal and submitting the interim report, etc.).
- Make appropriate prior arrangements for the demonstration.

Important: unauthorised late submission or failure to attend the demo/presentation could result in failure from the whole project.

The Supervisor

Supervisors complete the project sign-up process on the Moodle [Project Allocation System](#) (PAS) ensuring that the project topics are appropriate considering projects based solely on literature review activity and/or user/market surveys are not acceptable. Supervisors confirm the appropriateness of a project after the project proposal submission. You will be working with a supervisor with whom you will have the most contact throughout the year. The supervisor will provide appropriate guidance to the student working on the agreed project. A supervision meeting (project-related discussion/communication) can take place face-to-face or via any other appropriate medium, such as MS Teams, email. Students are entitled to 5 hours of supervision time in total throughout the year, with it being up to the student to arrange the meetings (or initiate the discussions) in the first instance. This supervision time is the absolute minimum and often, supervisors allocate more time than that for each one of their project students. Students can ask for meetings of various duration which would typically be 10-30 minutes, however, the supervisor can reduce/increase the duration, refuse or suggest another time based on their availability. This is only a guide: if your supervisor responds to your queries via email, or takes a deliverable to read and provide feedback on (e.g. the literature review for a research project) then this is also considered as supervision time. Such responses may take a longer time and also the place of a face-to-face/online meeting. It is important to maintain regular contact with your supervisor, to keep them informed of your progress, and to discuss with them any queries or problems that arise. The supervisor will provide formative feedback as appropriate, e.g., as an outcome of each supervision meeting, or after your submission of the project proposal and plan. The supervisor is the first marker of your project, who will give you feedback justifying the mark you receive for each assessed component of your project.

The Project Panel

Your final dissertation will be reviewed independently by three members of the Project Panel, apart from your supervisor. Your demonstration will also be independently marked by a chosen panel member.

The Project Coordinator

The Project Coordinator is a member of the academic staff responsible for the overall administration of UG projects; currently, this is Prof Ender Özcan. He will organise the project briefings and support lectures, maintain the Project Handbook and administer/coordinate the efforts of supervisors with respect to the sourcing, allocation, and assessment of projects, chair the panel meetings. The support lectures cover administrative details as well as aspects such as choosing, defining, and planning your project, project management, writing a dissertation, etc. See the module timetable for dates, times, and rooms.

Note that once you have a supervisor, queries concerning your project should always be directed to your supervisor first, not to the Project Coordinator.

Choosing a Project and Supervisor

You need to find a supervisor and agree on your project topic by the deadline (see deliverables and dates). The sooner you do this, the more choice you will have, as supervisor places fill up and you may not be able to find a supervisor who is willing to supervise the project you want to work on. A list of supervisors and project topics that they offer is given on the Moodle [Project Allocation System](#) (PAS). You can also come up with your own project topic, but you need to discuss it with your supervisor to make sure it is appropriate and the supervisor is happy to supervise it. Please ask your supervisor to sign you up using PAS.

If there are issues with PAS or you have a question, please contact the Project Coordinator.

Projects must include the students undertaking practical work of some sort using computing/IT technology. This is most frequently achieved by the creation of an artifact as the focus for covering all or part of an implementation life-cycle. Projects based solely on literature review activity and/or user/market surveys are not acceptable. Your supervisor will be checking and confirming that the project is of a suitable nature looking at your project proposal submission.

Transferable Skills

Final Year (3rd year) projects are intended to develop your transferable skills alongside your computing skills.

Transferable Skills are about developing generic skills which will be invaluable for your career. In particular, these skills include developing clear and logical thinking, analytical and problem-solving ability, communicating complex ideas and information as well as project management skills, ability to produce well-written reports and be confident in public speaking, good awareness of business and professional issues relevant to computing.

Laws, Social, Ethical and Professional Issues (LSEPI) in Projects

Your project doesn't happen in a vacuum. Your work is affected by the world around you. This influence includes laws and University policies, but also the concerns and priorities of society and your peers. On the other hand, your work can affect the world around you. In fact, it might also change the world (for better or worse)! The checklist, below, identifies a number of issues. Depending on the project, some of them may be more relevant than others. But every project student should have something to reflect on.

“Big picture” Reflection Checklist

Intellectual property (The Copyright, Designs and Patents Act (1988)):

- Has the work created any intellectual property? E.g. source code, documents, photos or recordings, patentable technologies, trademarks
- If so, what intellectual property and who owns it?
- Do you have a plan to commercially exploit the work? How?
- Do you plan to distribute or promote the work in other ways, for example using open source/open data/creative commons licenses? How?

- **See also** “Intellectual Property Rights” section below and [“Provision and Processing of Intellectual Property Rights for Students and Graduates at The University of Nottingham”](#)

Research ethics:

- Does the work involve human participants or data subjects?
- If so, have you followed the University of Nottingham research ethics requirements?
- How does the work safeguard participants and data subjects?
- How does the work reflect principles of informed consent and voluntary participation?
- Were there any ethically tricky aspects to the work, and how did you deal with them?
- **See also** the Research Ethics - Ethics Approval section above and [“Computer Science Research Ethics Resources”](#)

Data protection (The Data Protection Act (2018)):

- Is the work collecting or using personal data?
- If so, how is it complying with the requirements of data protection legislation?
- **Note**, this should also have been covered by your Research Ethics

Other legislation:

- Is the work affected by other legislation? If so, how is the legislation relevant and how has the work responded to that legislation? For example,
- the Equality Act (2010) has implications for accessibility,
- the Computer Misuse Act (1990) and Counter-Terrorism and Security Act (2015) have implication for (potentially) criminal use,
- the Online Safety Bill (if/when it comes into force) will have particular implications for systems accessed by children and other vulnerable groups,
- the Privacy and Electronic Communications (EC Directive) Regulations (2003) have implications (together with the Data Protection Act) for communication systems.

Broader ethical and social considerations:

- How have you considered the broader ethical and social aspects of the work?
- For example:
- Who will benefit from the work? How?
- Could the work have unintended consequences? What?
- Could the work have military or criminal applications? What?
- Who might be affected by the work, directly or indirectly? How?
- Have people who might be affected been able to influence the work?
- Is the work accessible or beneficial to all?
- Is the work likely to reduce or increase societal inequalities?
- What impact might the work have on the natural environment?
- Is the work worth doing? Why?
- **See** Responsible (Research and) Innovation (COMP3020). The [RRI Prompts and Practice cards](#), and the [Moral-IT cards](#) may also help you to reflect on these aspects of your project.

United Nations sustainable development goals:

United Nations (UN) calls all countries for an urgent collaborative action, setting 17 [Sustainable Development Goals](#) (SDGs) to serve as a "shared blueprint for peace and prosperity for people and the planet, now and into the future.":

1. No poverty: End poverty in all its forms everywhere
2. Zero hunger: End hunger, achieve food security and improved nutrition and promote sustainable agriculture
3. Good health and well-being: Ensure healthy lives and promote well-being for all at all ages
4. Quality education: Ensure inclusive and equitable quality education and promote lifelong learning opportunities for all
5. Gender equality: Achieve gender equality and empower all women and girls
6. Clean water and sanitation: Ensure availability and sustainable management of water and sanitation for all
7. Affordable and clean energy: Ensure access to affordable, reliable, sustainable and modern energy for all
8. Decent work and economic growth: Promote sustained, inclusive and sustainable economic growth, full and productive employment and decent work for all
9. Industry, innovation and infrastructure: Build resilient infrastructure, promote inclusive and sustainable industrialization and foster innovation
10. Reduced inequalities: Reduce inequality within and among countries
11. Sustainable cities and communities: Make cities and human settlements inclusive, safe, resilient and sustainable
12. Responsible consumption and production: Ensure sustainable consumption and production patterns
13. Climate action: Take urgent action to combat climate change and its impacts
14. Life below water: Conserve and sustainably use the oceans, seas and marine resources for sustainable development
15. Life on land: Protect, restore and promote sustainable use of terrestrial ecosystems, sustainably manage forests, combat desertification, and halt and reverse land degradation and halt biodiversity loss
16. Peace, justice, and strong institutions: Promote peaceful and inclusive societies for sustainable development, provide access to justice for all and build effective, accountable and inclusive institutions at all levels
17. Partnerships for the goals: Strengthen the means of implementation and revitalize the Global Partnership for Sustainable Development

Provide a discussion on how your project relates to or addresses any of the SDGs.

Intellectual Property Rights

An undergraduate student on a teaching course will normally have limited opportunity to create IP relating to their studies. However, on Computer Science courses the student may be involved in project work. Depending on the circumstances, the project work may involve working with other students, academic staff or other employees of the University. In these conditions IP created may be jointly owned between the collaborating parties. Many of the research groups are funded by research grants or industrial sponsorship with an expectation that arising IP will be protected and where possible commercially exploited. Consequently, the student may be requested to sign an agreement relating to confidentiality and IP prior to being allowed to work with the research group. The IP & Commercialisation team will set up the short confidentiality and IP agreement based on a standard template and can answer any queries regarding the

agreement which a student might have. The confidentiality provisions will not affect the ability of the student to submit a written project report and in the event that the student has contributed to IP which is later commercially exploited they will receive the same revenue share as they would if they were an employee of the University.

For more details, please refer to the "[Provision and Processing of Intellectual Property Rights for Students and Graduates at The University of Nottingham](#)" document. In order to discuss your specific situation, please contact the IP management group via tto@nottingham.ac.uk by sending an email with the subject line of "Computer Science: Final Year L3 project IP query".

Important Dates and Deliverables

The following list provides provisional important dates for the project students, including approximate deadlines for the submission of deliverables. Please refer to the Moodle COMP3003 project webpage for the exact dates when the academic year starts.

- **Early September:** Moodle [Project Allocation System](#) (PAS) opens for student project sign-up before the academic year starts.
- **Early October:** Deadline for students to agree on a project topic with a supervisor via Moodle [Project Allocation System](#) (PAS). Students who have not found a supervisor by this date will be assigned to available supervisors by the project coordinator.
- **Mid-October:** Deadline for submitting your detailed project proposal including a detailed project plan (up to four A4 sides) to your supervisor. Your supervisor will give you formative feedback on your project proposal and plan; you must revise it according to your supervisor's feedback.
- **Late October/Early November:** Deadline for submitting your
 - final project proposal (in PDF) via Moodle for supervisor approval, and
 - preliminary [Research Ethics](#) checklist after discussing the ethical issues with your supervisor in relation to your project via the Moodle project webpage. Should the preliminary checklist indicate that full ethics clearance is required; the full clearance process can be completed later if appropriate. However, note that **no** work requiring ethics clearance must be undertaken prior to having obtained clearance.
- **Around Christmas:** Deadline for submitting your interim report (in PDF). The interim report comprises **10%** of the mark for the module.
- **Around Easter (Following Year):** Deadline for electronic submission of your dissertation. **Dissertations will be marked based on electronic submissions.** Details of the suggested dissertation structure can be found in the Dissertation section and dissertation marking criteria in the Assessment section. The dissertation comprises **75%** of the mark for the module.
- **Just Before the Assessments Period (Following Year):** Final Presentation/Demonstration Days. You will give a final presentation/demonstration of your project. Assessment is based on the quality of work (e.g., software) and the presentation and communication. The demonstration/presentation mark comprises **15%** of the mark for the module.
- **In the Assessments Period (Following Year):** Release of demo marks.
- **After the Assessments Period (Following Year):** Release of provisional dissertation marks.

Project Proposal

The project proposals will not be assessed, however, they will be taken into account in the assessment of the interim reports. The project proposals should not exceed four A4 sides (2000 words) excluding the cover page, and containing the following main components:

- **Cover/title page:** This should include the project title, your name, ID, email, and name of the programme of your study (e.g., CS with AI).
- **Background and motivation:** Briefly describe the background to the project, importance/need of the area, and motivation for carrying out the proposed work.
- **Aim(s) and Objectives:** The aim is usually a single sentence describing at a high level what the point of the project is and what will be achieved. The objectives are sub-components of the general aim, detailing the individual aspects which need to be achieved in order to deliver the aim(s).
- **Work plan:** Describing the tasks to be carried out. The time plan should be realistic and should take into account other commitments such as exam periods, holidays, etc. The inclusion of a Gantt chart is strongly recommended as a visual representing the project schedule.
- **Bibliography** containing some key publications that are either explicitly referred to in the text.

Examples of Good Proposals

- [Vehicle Hot Spot Identification on Data Streams, Rebecca Tickle \[PDF\]](#)
- [The Swift Instant Messenger, Marcin Wisniewski \[PDF\]](#)
- [The Impact of Social Settings and Virtual Humans on Facial Expressions, In Response to Taste Stimuli, Humaira_Ahmed \[PDF\]](#)
- [Monadic streams library, Chris Purdy \[PDF\]](#)

Interim Report

An interim report consists of an outline with a progress update and draft content. The following is a typical interim report structure provided as guidance and this varies depending on the nature of the project. A brief description of each component can be found in the Dissertation section below. Please discuss your interim report with your supervisor before submission.

- **Introduction**
- **Motivation**
- **Related work**
- **Description of the work**
- **Methodology**
- **Design**
- **Implementation (if there is any)**
- **Progress**
 - **Project management** covering the tasks as a part of your **work plan as presented in your final proposal** and progress as well as how time and resources are managed. The need for reviewing the tasks and explanation of/reasons for any adjustment made to the future work plan. Inclusion of a Gantt chart is strongly recommended as a visual indicating the progress of the project.
 - **Contributions and reflections** providing the details of your achievements and contributions up to date as well as a personal reflection on the plan and your experience of the project (a critical appraisal of how the project has been

progressing). This section should also explain how you have considered the “bigger picture” within which your work is situated in terms of **LSEPI** (refer to the Laws, Social, Ethical and Professional Issues in Projects section above), highlighting which issues are the most significant for your project, and how you will address them in the remainder of your project.

- **Bibliography**

There is a **suggested** maximum limit of 5000 words without exceeding 15 pages (A4 sides) for the interim report, including the bibliography and excludes cover/front pages (e.g., abstract, acknowledgment, table of contents).

Examples of Good Interim Reports

The following samples are the main bodies of "good" interim reports selected from various years:

- [Predicting Solar Irradiance, Timothy Cargan \[PDF\]](#)
- [Scheduling and Optimisation of IT Support Incidents, Aidan Reed \[PDF\]](#)
- [Machine Learning for Market Prediction, Jay Mehta \[PDF\]](#)
- [A Data-Driven Approach to Bus Timetable Optimisation Recommendations, Jonathan Foot \[PDF\]](#)
- [Novel Approaches for Optimal Control of Battery Energy Storage Systems, Eskil Sulen Gjerde \[PDF\]](#)
- [Improving Selection Hyper-Heuristics Based on Analyses of their Search Behaviour using HyVis, Jack Cakebread \[PDF\]](#)
- [Predicting Key Presses over Video Calls using an Audio Side-Channel Attack and Machine Learning, Adam Masters \[PDF\]](#)

Dissertation

A demo video needs to be submitted as a part of the dissertation, which will also get assessed.

The following is a typical dissertation structure. However, this is for guidance only: depending on the type and specific aspects of your project, you may find it appropriate to structure your dissertation in a more or less different way. Discuss with your supervisor.

- **Title page** with a signed declaration that the dissertation is your own work [[WORD template](#)]. If submitting electronically, your initials would suffice as a signature. You are reminded of the University's guidelines on [Academic Integrity and Misconduct](#).
- **Abstract** giving a short overview of the work in your project.
- **Table of contents** giving page numbers for all major section headings.
- **Introduction** setting out the aims and objectives of your project, **explaining the overall intention of the project and specific steps that will be taken to achieve that intention.**
- **Motivation** explaining the problem being solved and its importance or need for it.
- **Related work** explaining what your project does that is new or is better than existing work in the same field
- **Description of the work** explaining what your project is meant to achieve, how it is meant to function, e.g., perhaps even a functional specification for a software-oriented project

- **Methodology** describing the (theoretical/experimental/analytical/numerical/software development-based/research-based, etc.) methodologies/techniques/tools/algorithms/technologies/etc. that are relevant to the project topic justifying the choices made for the project, where possible with supporting evidence derived from the existing work.
- **Design** containing a comprehensive description of the design chosen, how it addresses the problem, and why it is designed the way it is
- **Implementation** containing a comprehensive description of the implementation of your software, including the language(s) and platform chosen, problems encountered, any changes made to the design as a result of the implementation, etc.
- **Evaluation** explaining how your software/approach was tested (using different datasets or in different environments), statistical evaluation of performance, results of user evaluation questionnaires, etc.
- **Summary and Reflections** including a discussion of results in a wider context (considering other work).
 - **Project management covering** the tasks as a part of your work plan and progress as well as how time and resources are managed.
 - **Contributions and reflections** providing the details of your achievements and contributions including innovation, creativity and novelty (if there is any) as well as a personal reflection on the plan and your experience of the project (a critical appraisal of how the project went). This section should also explain how you have considered the “bigger picture” within which your work is situated in terms of **LSEPI** (refer to the **Laws, Social, Ethical and Professional Issues in Projects** section above). You should present your reflections on each issue, including why they are (or are not) significant for your project.
- **[NEW] Declaration of AI Usage** subsection **must be included** in all dissertations as a short paragraph immediately before the bibliography. Please refer to the **AI Use Policy** section above for the principles governing the acceptable use of AI tools. This declaration subsection is not included in the word limit. As this is a newly introduced requirement, previous dissertation samples may not contain examples of such statements, so here are some:
 - I confirm that this dissertation is my own original work. I have not used Artificial Intelligence (AI) tools to generate academic content or ideas.
 - No AI-generated content has been presented as my own original work.
 - AI tool (e.g., ChatGPT version...) was used to support with proofreading (e.g., grammar, spelling, vocabulary, and writing style) for the
 - whole dissertation, or specific sections (e.g. chapter 1, 2, and 4, ...)
 - AI tool (e.g., Claude version...) was used for identifying coding errors in the implementation of
 - algorithms (e.g., quick sort algorithm),
 - classes (e.g., class A),
 - procedures/functions (e.g., procedure B), etc.
 - AI tool (e.g., Copilot version...) was used for writing assistance including
 - improvement of illustrations diagrams/etc. (e.g., Figure 1-3 and Table 4-6,...)/
 - or similar non-content-generating support.
- **Bibliography** containing a *complete* list of books and other publications that are either explicitly referred to in the text, or which are recommended for further reading on the topic

- **Appendices**, e.g., User Manuals, supporting evidence for claims made in the main part of the dissertation (e.g. a copy of a user evaluation questionnaire), samples of test data, etc. Note that Appendices are optional

There is a maximum limit of 15,000 words without exceeding 40 pages (A4 sides) for the **main body** of the dissertation that will be submitted in PDF. This limitation includes the bibliography and excludes cover/front pages (e.g., abstract, acknowledgment, table of contents) and excludes the appendices, listing of any codes or any other supporting documentation.

Note: Your dissertation **should not exceed the word and page limits**. You do not have to use up your word limit to get a good grade; never 'pad out' your dissertation - which is likely to get penalised. The length of the report is one of the marking criteria that will be considered by the markers and you can always violate a requirement at your own risk, as you could lose marks.

Dissertation – Demo Video

You should prepare a **1 to 3-minute** long presentation and record it using a (screen capture) software tool as a demo video. The maximum limit for the duration of the demo video should not be exceeded. You can always violate a requirement at your own risk as you could lose marks. A demonstration video should be encoded and submitted as 1280x720 H264 video in an MP4 wrapper. This is a standard format that can be generated from most video software on all platforms. Videos submitted in other formats will be rejected. The focus of your demo video should be the outcome of your project. For example, for a software-oriented work, you should discuss the resultant software product illustrating its (main) features, how it works and your contribution in the demo video. For a research-oriented project, the novelty of the work, your contribution, and results should be discussed. As a general guide for the demo videos,

- prepare the narrative for the demo video before the recording
- you can edit pieces of narrative to produce the final demo video
- remember having the title of your project on the first video frame and **stating** the title of your project and your name (do not use or mention your ID).
- speak clearly
- use audio-visual aids as appropriate illustrating crucial features of your work.

The following sample demo videos are provided just as a guide.

- [Vehicle Hot Spot Identification on Big Data Streams \[MP4\]](#)
- [Multimodal Function Optimization using Differential Evolution \[MP4\]](#)
- [Creating a Unity Tool to Make Finite Spaces Infinitely Explorable with Overlapping Geometry \[MP4\]](#)
- [Novel Approaches for Optimal Control of Battery Energy Storage Systems \[MP4\]](#)

Examples of Good Dissertations

The following samples are the main bodies of "good" dissertations selected from various years:

- [A Study of Parallelisation of Pattern Search and Covariance Local Search, Yuyang Zhou \[PDF\]](#)
- [A feet prediction and correction system for a 3D character in Unity, Zain Ali \[PDF\]](#)
- [Computer-aided Expert System for Breast Cancer Diagnosis, Xin Yu Liew \[PDF\]](#)

- [Novel Approaches for Optimal Control of Battery Energy Storage Systems, Eskil Sulen Gjerde \[PDF\]](#)
- [Programming with Monsters, Christopher Purdy \[PDF\]](#)
- [Semantic segmentation and quantification of spontaneous intracerebral haemorrhage on CT images using deep learning, Yong En Kok \[PDF\]](#)
- [Fugue: A Friendly Functional Programming Language with Holes, Zachariah Garby \[PDF\]](#)

Dissertation Submission Instructions

Your dissertation needs to be submitted electronically via Moodle. You **do not need** to submit "large" library files for the code part of your submission; **only the source code** of your implementation (along with instructions regarding which libraries should be included in the environment for the execution of your code in a README file) would be sufficient if the overall size of the code exceeds the limit. Please follow the precise submission instructions on Moodle.

Your dissertation will be marked based on your electronic submission. Late submission (of any one component: the main body of the dissertation, demo video, supplementary material, and code) is subject to the standard University penalty of **5%** reduction per working day. For more information, please refer to the [Policy on Late Submission of Assessed Materials](#).

Electronic submission

You need to make your electronic submission in **four** parts. Please submit all files listed below using Moodle (under Assignments choose the link to Dissertation), **by the deadline**

(i) main body of your dissertation as a PDF (the only accepted format!). When you include a cover or front page, please provide only your ID and not your name/surname.

(ii) demo video: recording of the demo as an MP4 (the only accepted format!)

(iii) other supplementary material (such as, appendices, data, lengthy proofs, length empirical results or any other supporting documentation) as one file (.zip or .tar .gz)

(iv) your code (preferably the source code and a readme file providing links to the libraries for creating the executable that will be used during the demo, including any end-user documentation) as a compressed file (.zip, .jar or .tar.gz).

Please contact the Project Coordinator if you have problems accessing Moodle or have a good reason to request an extension. Your supervisor cannot give you an extension (neither can the module convenor without an approved extenuating circumstances form).

Demonstration

An essential component of the project is to present your work and discuss it in a professional manner. The duration of a demonstration/presentation is fixed and includes a question and answer session. The exact duration for demonstrations is announced via the Moodle project web page. Depending on the nature of your project (e.g, research or software), an appropriate combination of presentation/demonstration would be required. Please take advice from your

supervisor on this. The demonstration/presentation/viva is usually organised by the Project Coordinator and announced via the Moodle project web page. Normally, your supervisor and second marker who is a panel member attend the demonstration. Where relevant the external sponsor or anyone else the supervisor thinks is appropriate could attend.

Warning: If you do not attend your demonstration/presentation, you could fail the whole project.

The demonstration/presentation gives the markers of your project the chance to gather further information on the quality of your work (e.g., research, software product), as well as to observe how well you can demonstrate/present it. Considering the limited demonstration time, you should prepare carefully and do a mock-up demonstration/presentation before the date, perhaps with your supervisor or friends. You may need to be selective on what you are going to demonstrate/present, however, make sure that you cover the key objectives and outcomes of your project as well as your contributions and originality (if there is any) of your work. You need to make appropriate prior arrangements for the demonstration/presentation. As general guidance for the demonstration/presentation,

- remember introducing yourself
- be enthusiastic about your own work – you worked hard for a year to achieve the project goals
- speak clearly making eye contact with the *audience*
- use visual aids as appropriate
- be ready to explain the limitations of your work, difficulties that you encountered, and what further work could be done.

Assessment

Projects will be assessed based on the following aspects of your project:

- Knowledge and understanding
- Critical thinking skills
- Technical achievement
- Technical writing skills
- Professionalism

The marking criteria that will be used **as a guideline** in the assessment of each project component is explained below.

Interim Report Marking Criteria

The interim report is marked by the supervisor only **[10%]**. Considering that the work is in progress, a subset of the relevant dissertation marking criteria (e.g., technical writing skills, appropriateness of the aim and objectives, knowledge on the subject area and topic, critical thinking skills, progress and plan) will be used in the assessment. Evidence should be available of substantial work having taken place, leading to good progress towards the end of the semester against the project plan the students were to have handed in by the deadline. Any reason for being behind plan should be explained through good reflective reasoning – not just I did not work hard enough.

More details on the marking criteria can be found in the **interim report marking rubric** [\[PDF\]](#). This rubric provides a set of guidelines outlining the typical requirements of specific grade band for the interim reports. It is not necessary that a project exhibit all the criteria listed in each grade given below to fall within that band. The characteristics of each band are merely representative of that grade. The balance between various aspects of the particular work will be considered during assessment. The supervisor will provide a written feedback justifying the final mark.

Demonstration Marking Criteria

The demonstration is marked by the supervisor **[15%]** and project panel member based on two main criteria:

- **[10%]**: quality of work (e.g., quality of software, degree of originality, and individual contribution), and
- **[5%]**: quality of presentation and communication (e.g., time management, quality of content, degree of clarity, quality of structure and style, ability to make the demo interesting, ability to respond to questions and discuss).

The supervisor will provide agreed written feedback justifying the given mark.

Dissertation Marking Criteria

Your supervisor will grade your dissertation based on the following criteria and this will be moderated by a panel of markers **[75%]**.

- Knowledge and understanding
 - Breadth of knowledge on the subject area and knowledge of the chosen specific topic as the focus
 - Understanding of the (software development/research) methodologies / techniques / tools related to the project (e.g., specification, design, synthesis, etc.)
 - Quality & usage of sources
- Critical thinking skills
 - Extent of the justification of the method(s) (theoretical / experimental / analytical / numerical / design / objectives). E.g. related to theory / literature / standard practices / performance requirements / specification
 - Ability to describe relevant results
 - Discussion and interpretation of result/ evaluation of software (own project): ability to sum up and interpret them, showing an awareness of limitations of own work.
 - Discussion and interpretation of results/ evaluation of software in wider context: ability to interpret own results in wider context.
 - Reflection on original plan
- Technical achievement
 - Quality of software (work/product/ implementation/ research)
 - Quality of supporting documentation (e.g., testing results, input/output data, user manual, comments etc.)
 - Attainment of agreed appropriate objectives and requirements
- Technical writing skills

- Section headings
- Visual elements (graphs, figures, tables): range of types of visual elements, quality and adequate presentation (labels, captions)
- Degree of consistency of the formatting of the document (font size, references list, use of bold/italic, layout, spacing, etc.)
- Structure of sentences and division of text into paragraphs
- Use of in-text citations to support development of argument
- Accuracy of the grammar
- Typos
- Style (use of technical vocabulary)
- Length of report
- Professionalism
 - Ability to self-manage
 - Professional approach to project and time-management
 - Innovation and specific individual contributions
 - Demonstration video

More details on the marking criteria can be found in the project **dissertation marking rubric** [\[PDF\]](#). This rubric provides a set of guidelines outlining the typical requirements of a specific grade band for the dissertations. It is not necessary that a project exhibit all the criteria listed in each grade given below to fall within that band. The characteristics of each band are merely representative of that grade. The balance between various aspects of the particular work will be considered during assessment. The supervisor will provide written feedback justifying the final mark.

Resits

Resit projects get assessed based on 80% dissertation and 20% demonstration. The supervisor acts as the first marker for assessing the dissertation and demonstration components, while a second marker is also assigned for marking moderation. The same deadline for the dissertation submission will be followed in the same academic year when the resit takes place. Depending on the circumstances, there is flexibility to arrange "remote"/online demonstrations over an appropriate medium, such as MS Teams.